

ECEN721: Optical Interconnects Circuits and Systems Spring 2026

Lecture 8: VCSEL TXs



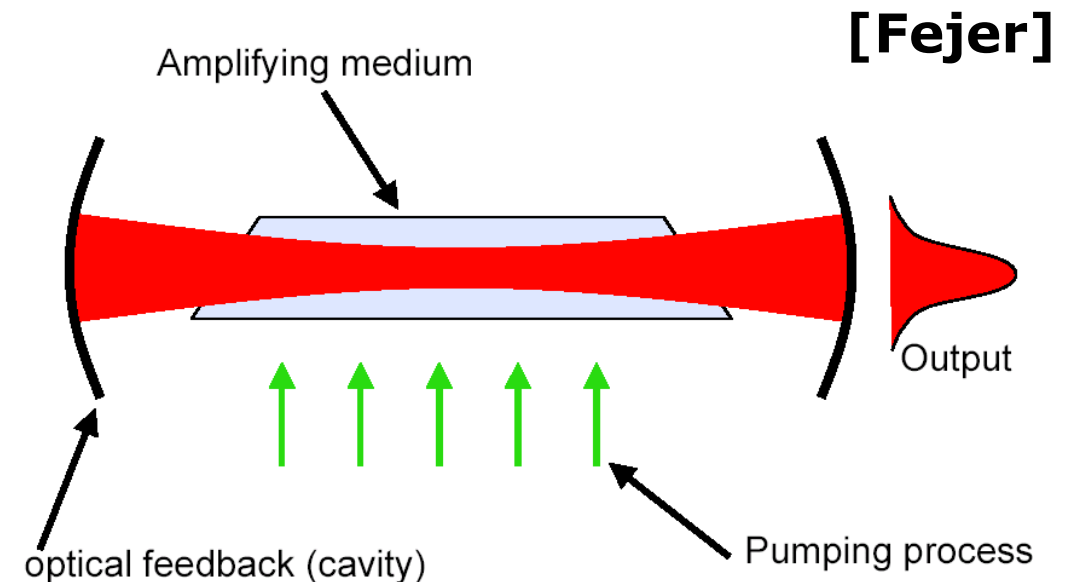
Sam Palermo
Analog & Mixed-Signal Group
Texas A&M University

Announcements

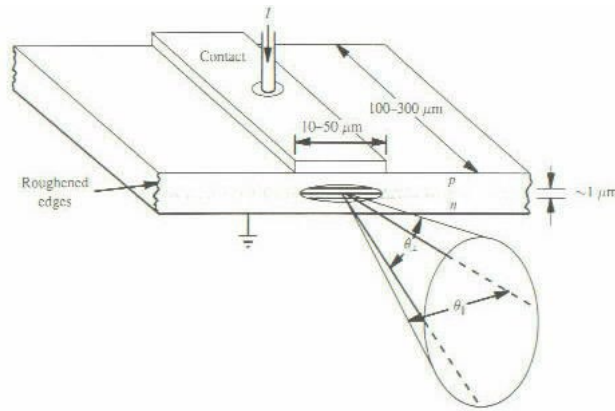
- Exam 1 Mar 3
 - In class
 - One double-sided 8.5x11 notes page allowed
 - Bring your calculator
 - Covers through Lecture 6
- Homework 3 due Mar 19

What is a Laser?

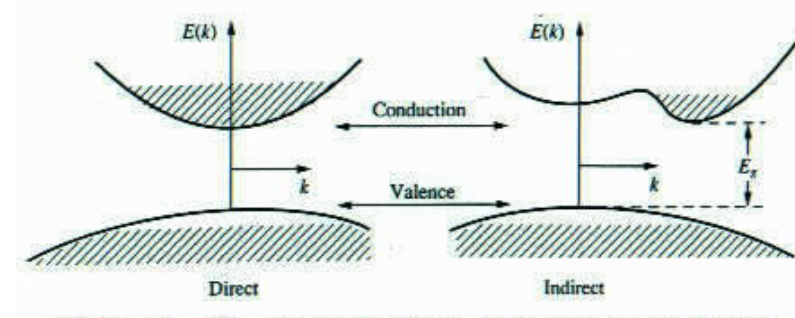
- Light **A**mplification by **S**timulated **E**mission of **R**adiation
- Light **O**scillation by **S**timulated **E**mission of **R**adiation
- Lasers are optical oscillators that emit coherent light through the process of stimulated emission
- 3 Elements in all lasers
 - Amplifying Medium
 - Pumping Process
 - Optical Feedback (Cavity)



Semiconductor Diode Lasers



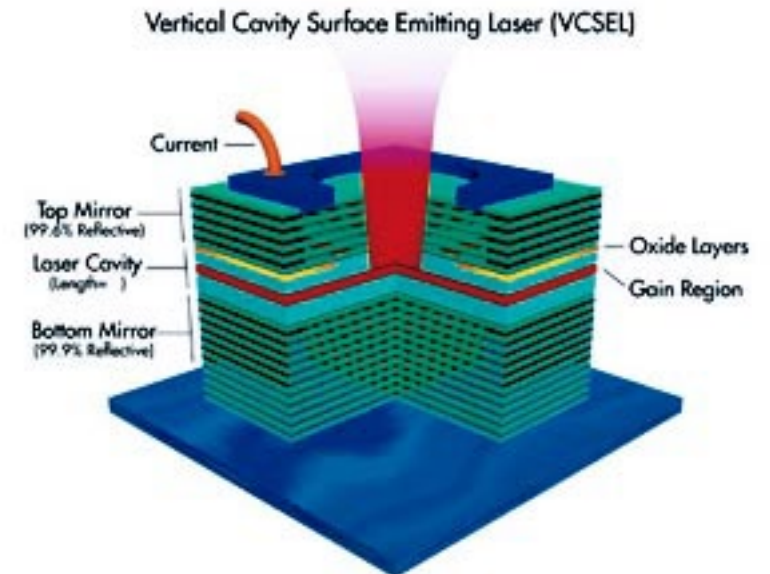
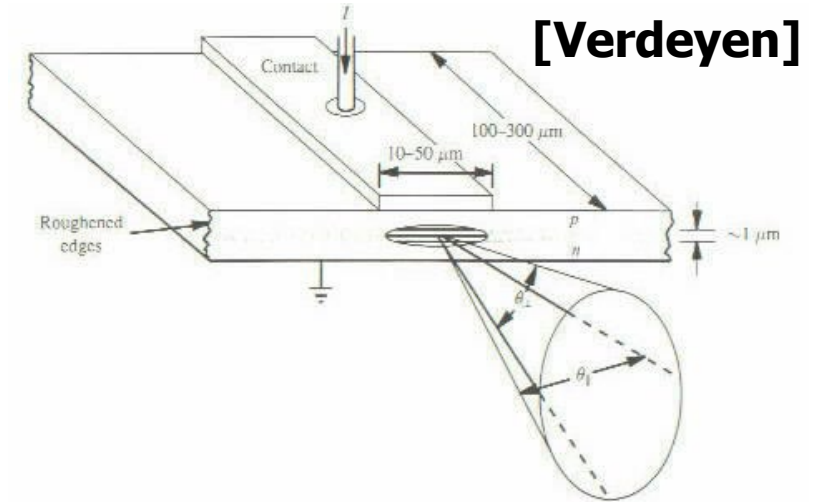
[Verdeyen]



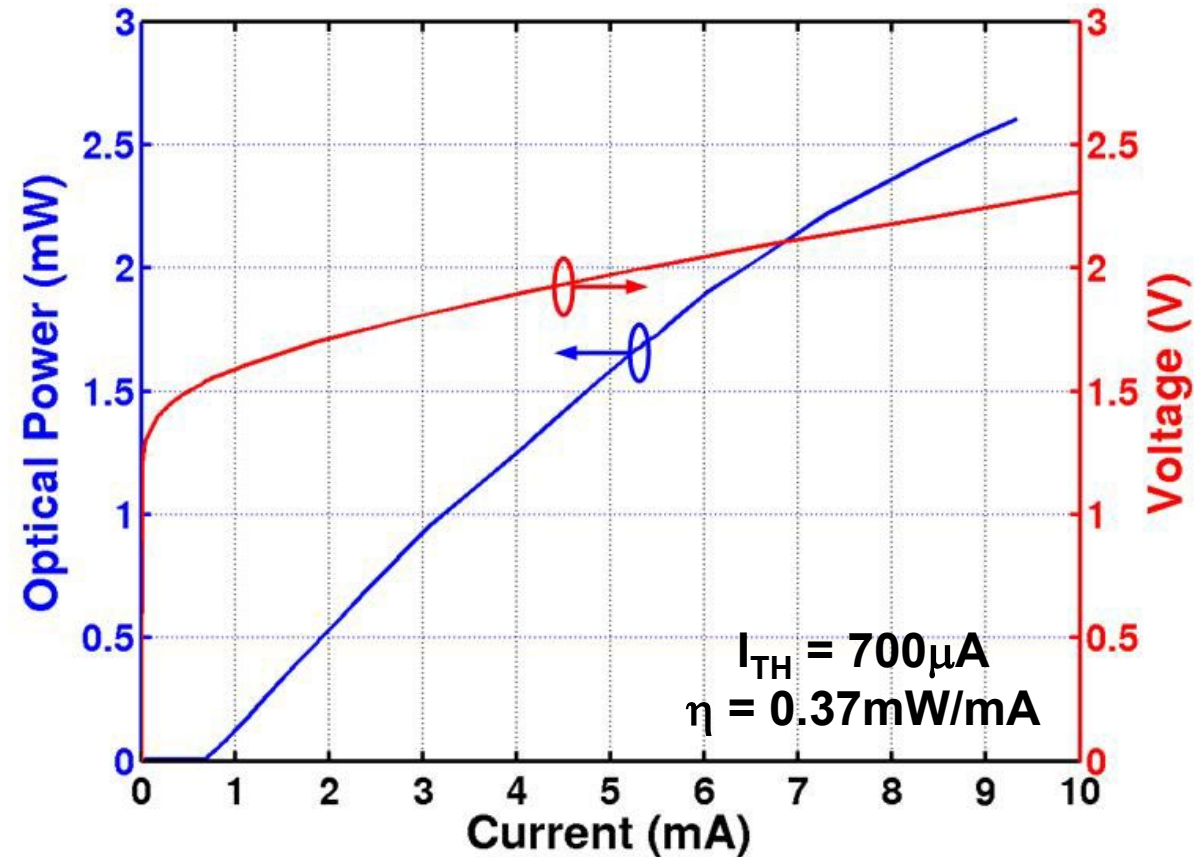
- Can be made with simple p-n junction
- Based on transitions between bands
 - Direct bandgap materials necessary
 - Si isn't $\Rightarrow$ GaAs, InP
- Pumped electrically with current source
- Efficient device requires confinement of both carriers and photons
 - Leads to the use of heterostructures

Edge Emitters & VCSELs

- Edge Emitters
 - Advantage
 - Historically easier to manufacture
 - Disadvantages
 - Emit light in an elliptical mode
 - Higher testing and packaging costs
- VCSELs – Vertical Cavity Surface Emitting Lasers
 - Advantages
 - Can make 2-D arrays
 - Emit light in a circular output mode
 - Smaller device $\Rightarrow$ Lower operating currents
 - Lower testing and packaging costs
 - Disadvantage
 - Hard to manufacture due to growth of high reflective mirrors

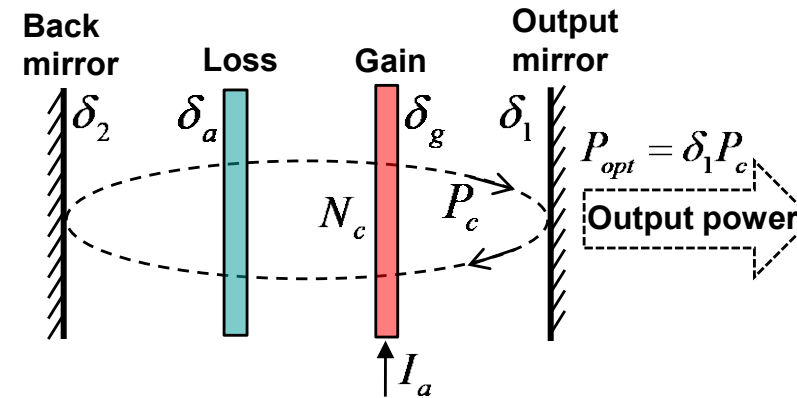
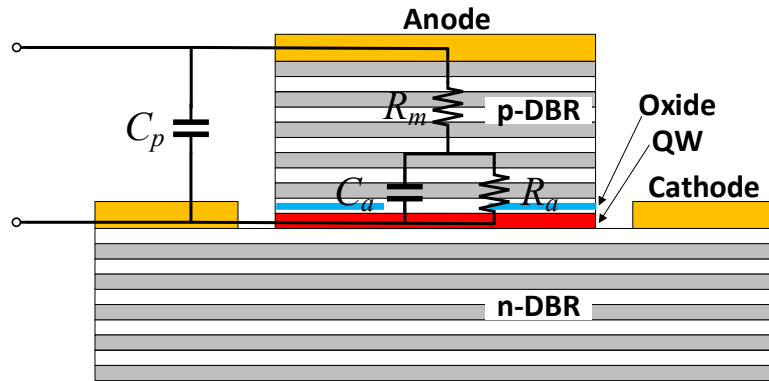


VCSEL Light-Current-Voltage (LIV) Curve

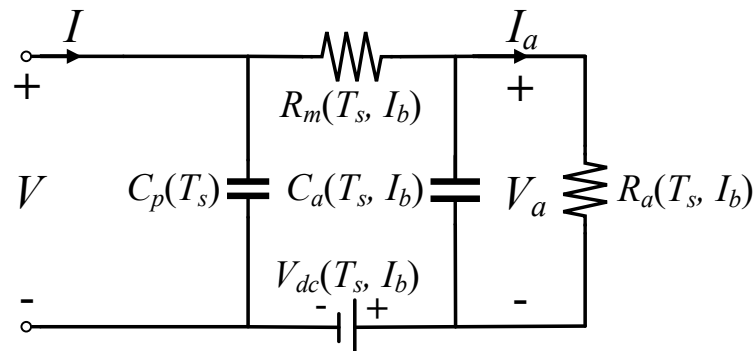


$$P_o = \eta(I - I_{TH}) \quad \text{Slope Efficiency } \eta = \frac{\Delta P}{\Delta I} \left(\frac{\text{W}}{\text{A}} \right)$$

VCSEL Model



Electrical Input Stage



Optical Output Stage

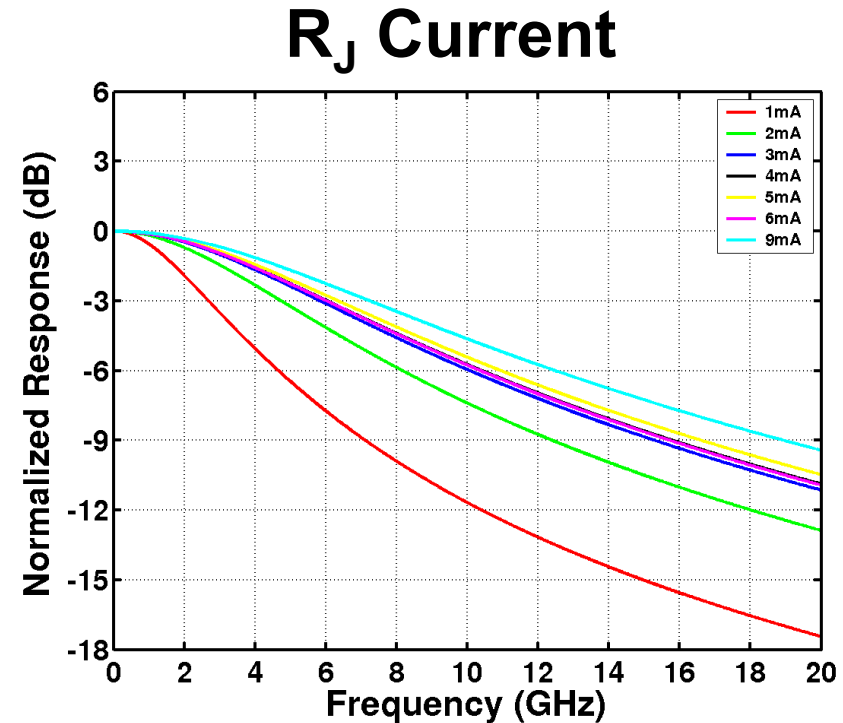
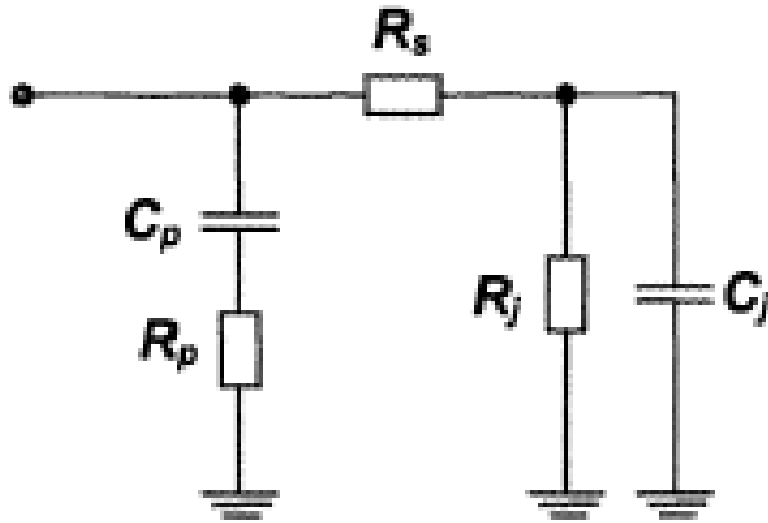
$$\frac{dN_c}{dt} = \frac{\eta_i(T) \cdot [I_a - I_{th}(T)]}{q} - \delta_g(T) \cdot \frac{P_c}{h\nu}$$

$$\frac{dP_c}{dt} = [\delta_g(T) - \delta_c] \cdot \frac{P_c}{\tau_c} + P_{sp}(T)$$

$$P_{opt} = \delta_1 P_c$$

- Capture thermally-dependent electrical and optical dynamics
- Provide dc, small signal, and large-signal simulation capabilities

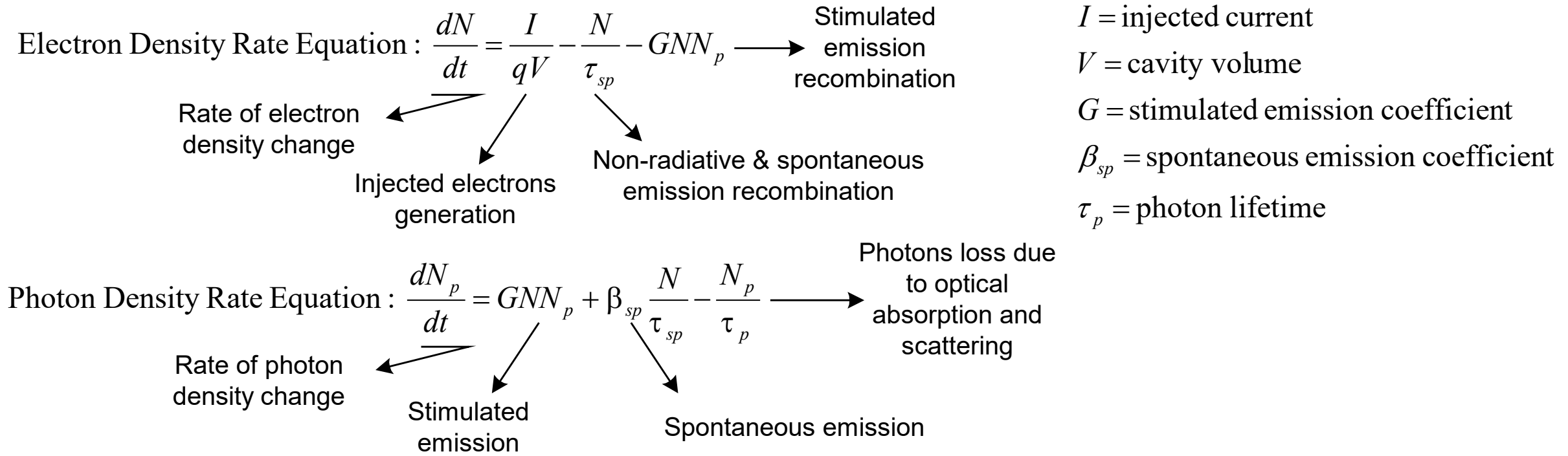
10Gbps VCSEL Electrical Model



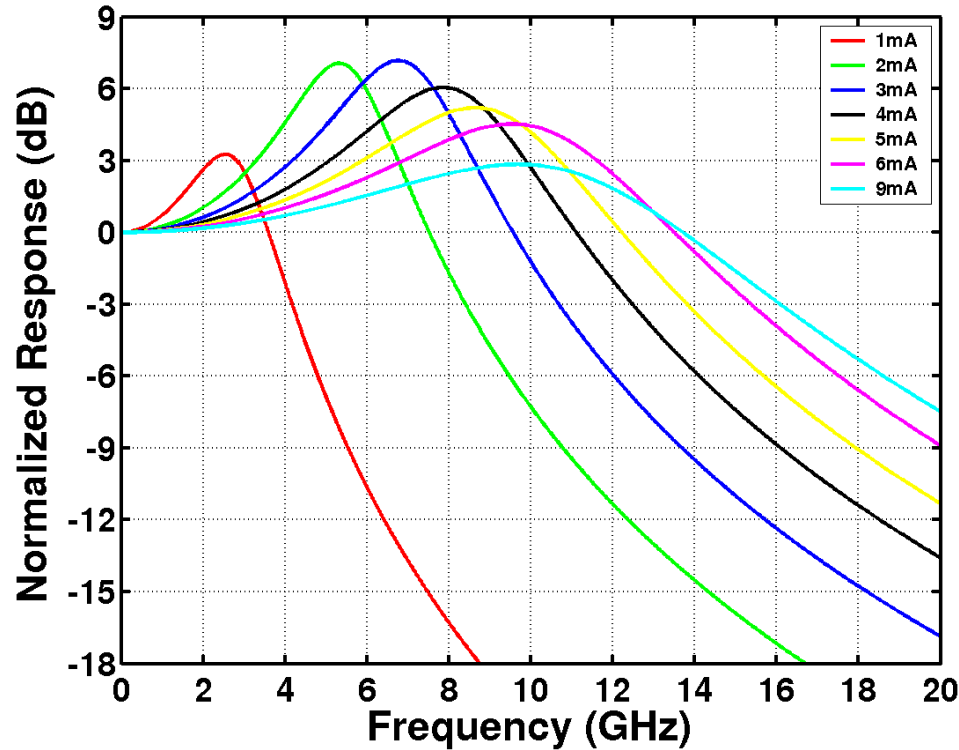
- Models finite Q pad capacitance, mirror series resistance, and junction RC
- Frequency response dominated by current dependent $R_j C_j$
 - f_{3dB} about 6.5GHz with $I \geq 3mA$

Laser Rate Equations

- Two coupled differential equations describe the electron density (N) and the photon density (N_p) interaction



VCSEL Rate Equation Frequency Response



$$\frac{P_{ac}(\omega)}{I_1(\omega)} = \frac{h\nu v_g \alpha_m}{q} \frac{GN_{pdc}}{-\omega^2 + j\omega \left(GN_{pdc} + \frac{1}{\tau_{sp}} \right) + \frac{GN_{pdc}}{\tau_p}}$$

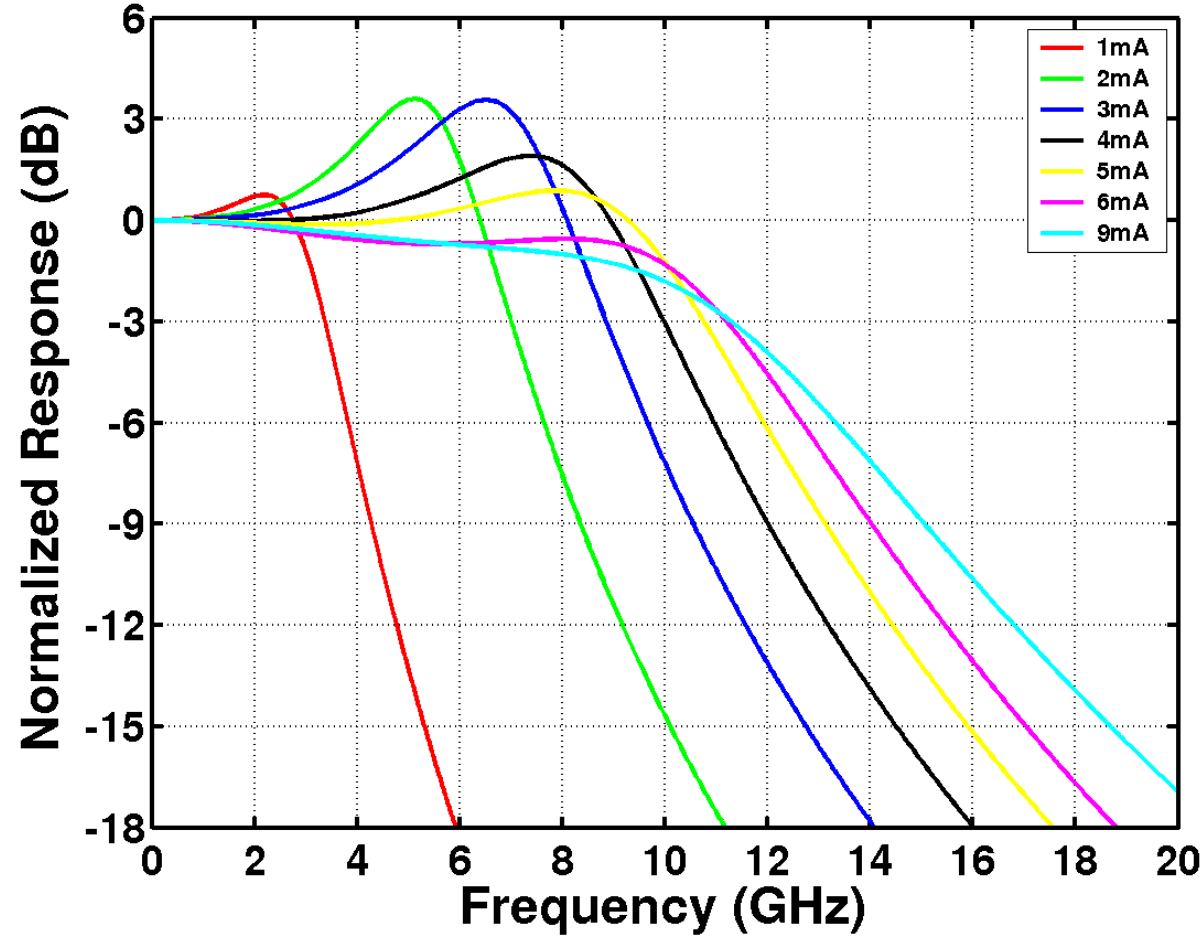
$$\omega_R = \sqrt{\frac{GN_{pdc}}{\tau_p}} \quad (\text{Proportional to } \sqrt{I_{avg}})$$

$$\zeta = \frac{GN_{pdc} + \frac{1}{\tau_{sp}}}{2\sqrt{\frac{GN_{pdc}}{\tau_p}}}$$

v_g = group velocity

α_m = mirror loss coefficient

10Gb/s VCSEL Frequency Response



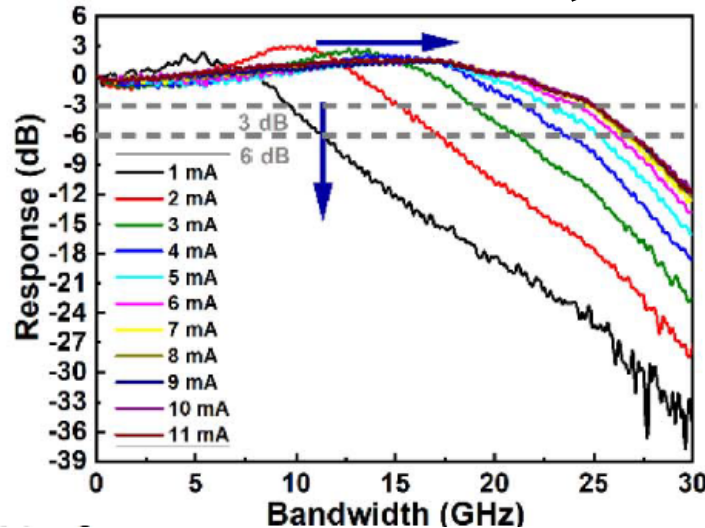
$$BW \propto \sqrt{I_{avg} - I_{TH}}$$

D. Bossert *et al*, "Production of high-speed oxide confined VCSEL arrays for datacom applications," *Proceedings of SPIE*, 2002.

VCSEL Bias-Dependent Bandwidth

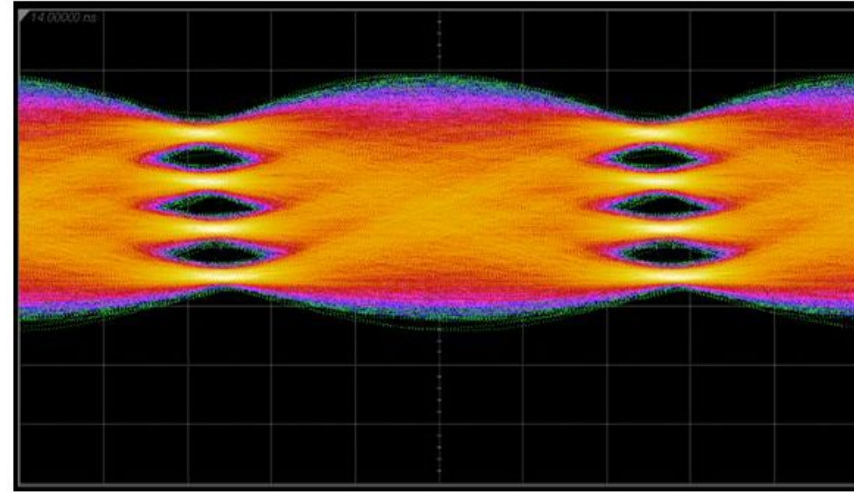
[Wu PhRes 2024]

$$BW \propto \sqrt{I_{avg} - I_{TH}}$$



200Gb/s 850nm MM-VCSEL
(c) 9mA, 50m (EMB 7500 MHz.km)

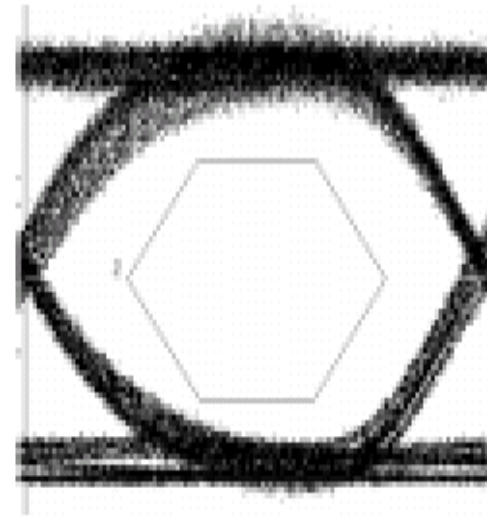
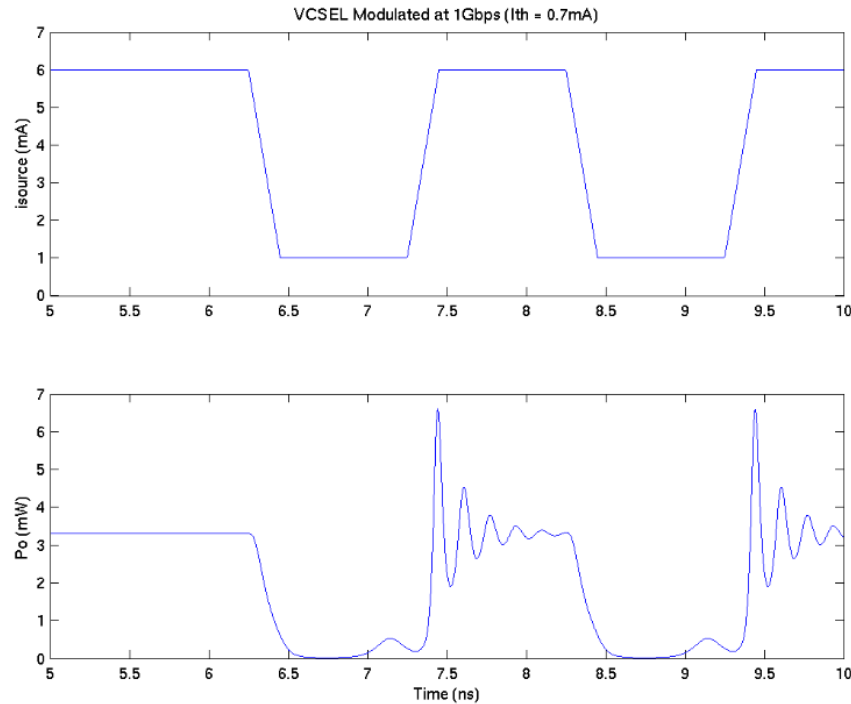
[Wang PhWest 2025]



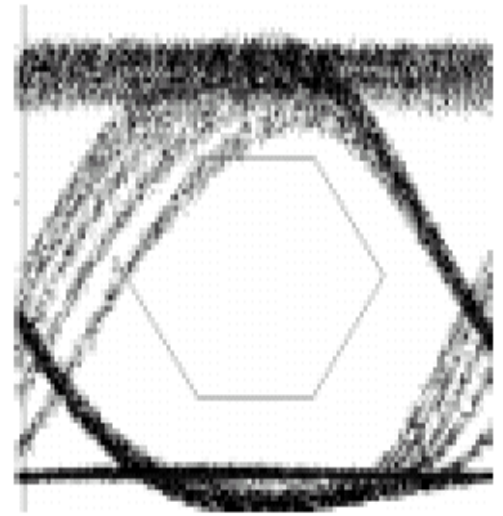
- 9mA bias
- 7-tap TX FIR
- 2.2dB ER
- 50m enhanced MMF
- 25-tap FFE at scope (RX)

- VCSEL electrical impedance and rate equations cause bias-dependent bandwidth
 - State of the art is 39GHz at 9mA bias [Wang PhWest 2025]
- Lower bias results in an underdamped low-bandwidth response
- Driving the VCSELs significantly above threshold reduces ER and results in a speed/power tradeoff
- 200Gb/s PAM4 demonstrated

Laser Rate Equations: Transient Response



Turn-on Delay



[Honeywell]

- Laser step response displays relaxation oscillations due to low damping
- Turn-on delay (t_d) occurs if the laser is biased below threshold
 - Causes data-dependent jitter

$$t_d = \tau_{sp} \ln \left[\frac{I_1 - I_0}{I_1 - I_{TH}} \right]$$

Chirp

- VCSELs also have additional unwanted frequency modulation called **chirp**
- The linewidth in this case can be approximated as

$$\Delta\lambda \approx \frac{\lambda^2}{c} B \sqrt{\alpha^2 + 1}$$

where α is the *chirp parameter* or *linewidth enhancement factor*.

- The α parameter relates the change in optical frequency to the change in optical power

$$\Delta f(t) \approx \frac{\alpha}{4\pi} \cdot \frac{d}{dt} \ln P_{out}(t)$$

- Directly modulated lasers have positive α values, implying that for a rising edge the laser will blue-shift (higher frequency/shorter λ) and red-shift for a falling edge

Relative Intensity Noise (RIN)

- VCSELs have occasional spontaneous emissions which add amplitude and phase noise to its coherent light output
- The resulting intensity fluctuations are known as **relative intensity noise (RIN)**
- At the receiver, this will get converted to an equivalent electrical noise component by the photodetector which is approximately proportional to the received signal power

$$\overline{i_{n,RIN}^2} = RIN \cdot I_{PIN}^2 \cdot BW_n$$

Here RIN is a parameter characterizing the laser RIN noise measured in dB/Hz. The resulting SNR is

$$SNR = \frac{I_{PIN}^2}{\overline{i_{n,RIN}^2}} = \frac{1}{RIN \cdot BW_n}$$

Can't improve SNR by increasing the laser power!

RIN Power Penalty

Assuming a laser with $RIN = -135dB/Hz$ and a 10GHz receiver noise bandwidth, the SNR is

$$SNR = \frac{I_{PIN}^2}{i_{n,RIN}^2} = \frac{1}{10^{\frac{-135dB/Hz}{10}} (10GHz)} = 3.16 \times 10^3 = 35dB$$

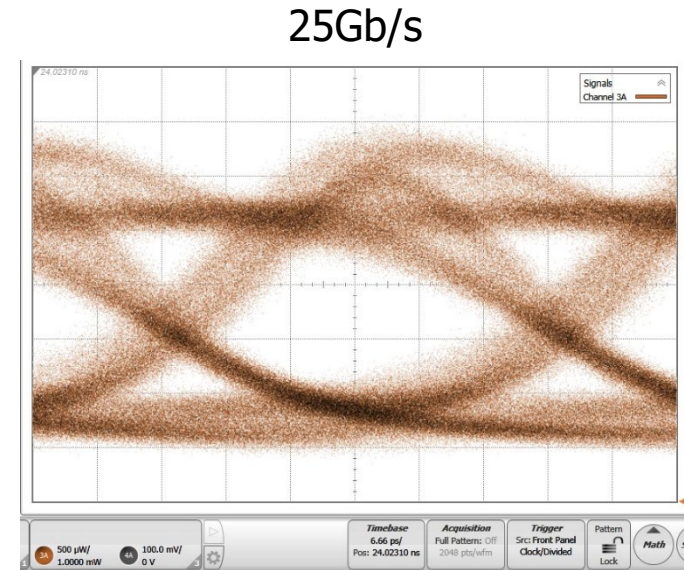
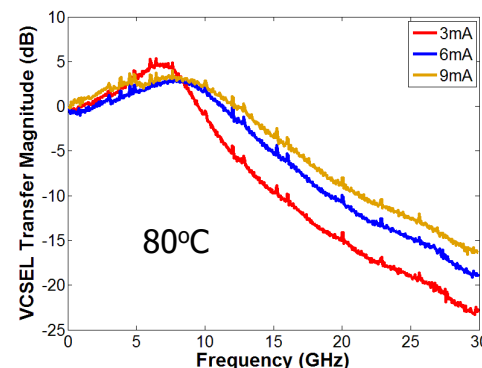
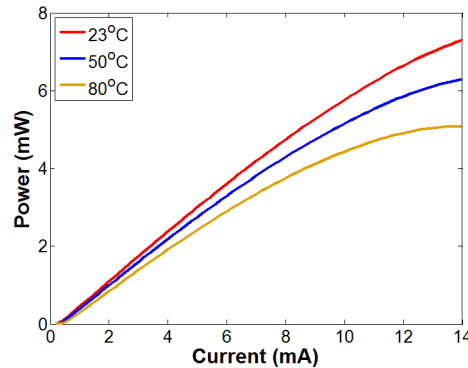
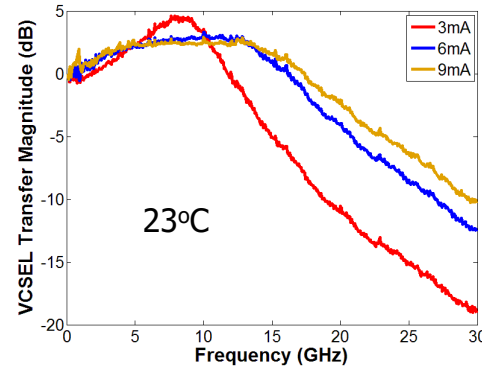
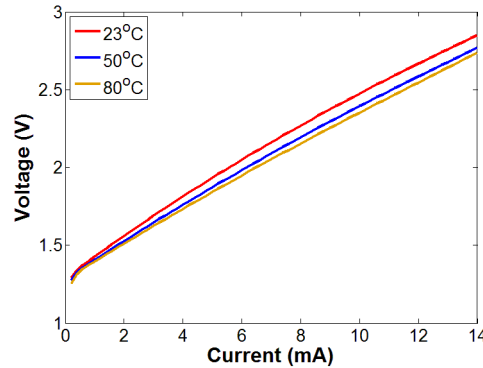
- This SNR is fine for digital NRZ signaling, but may be an issue for analog optical links for applications such as cable TV
- RIN noise does introduce an additional power penalty

$$PP = \frac{1}{1 - Q^2 \cdot RIN \cdot BW_n}$$

For the above example, the BER = 10^{-12} power penalty is

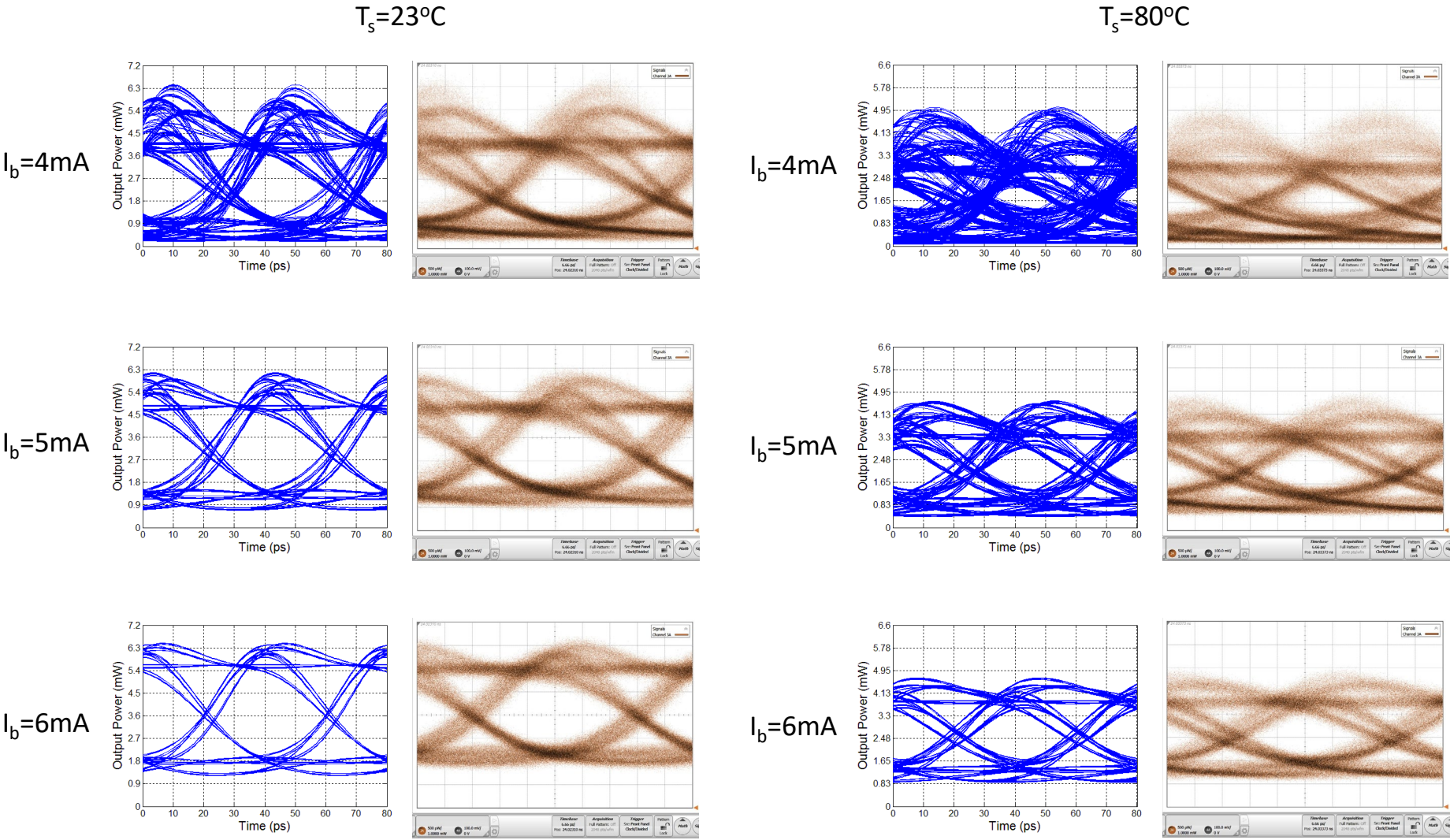
$$PP = \frac{1}{1 - (7.035)^2 \left(\frac{1}{3.16 \times 10^3} \right)} = 1.016 = 0.069dB$$

Temperature-Dependent Performance

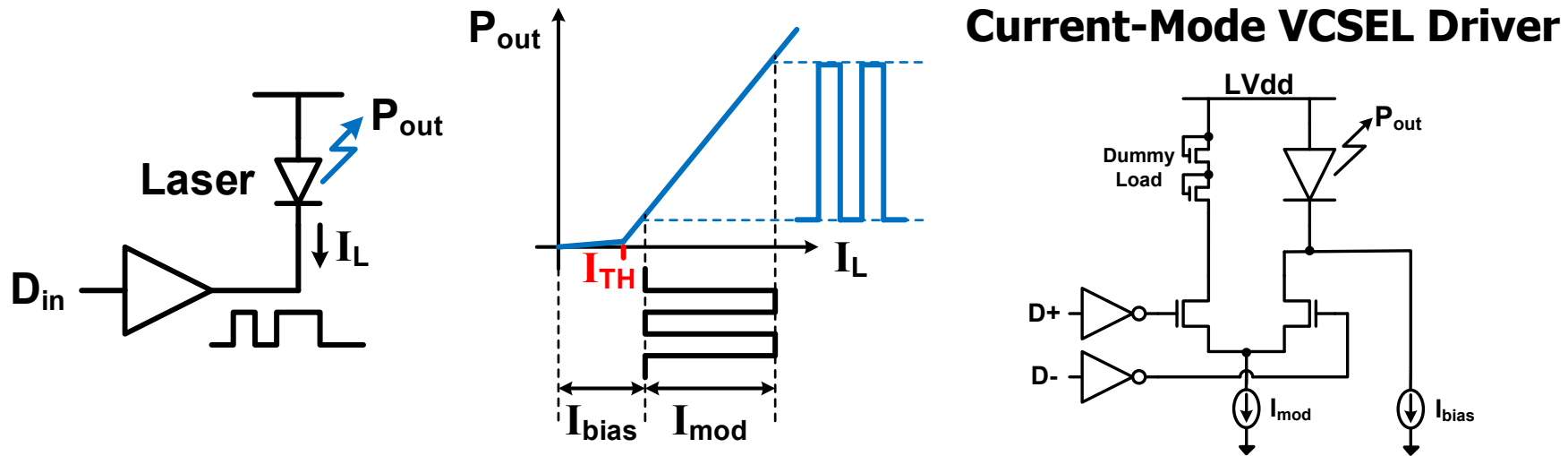


- Optical power-current-voltage (L-I-V) response is temperature-dependent
- Bandwidth is bias and temperature dependent

Measured & Simulated 25Gb/s Eye Diagrams

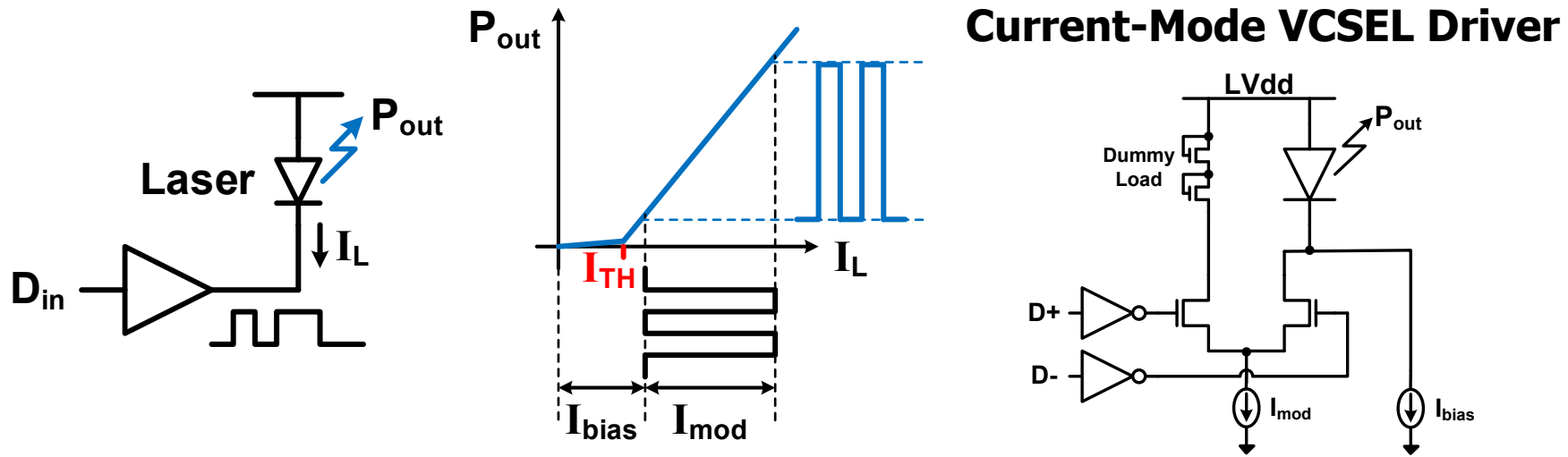


Laser Drivers



- Current-mode drivers are often used due to the laser's linear L-I relationship
- In addition to the high-speed modulation current I_{mod} , laser drivers must also supply a bias current I_{bias} to ensure a minimum frequency response and/or eliminate turn-on delay

Laser Drivers



- The total laser current depends on whether the high-speed modulation current is DC- or AC-coupled
- DC-coupled case

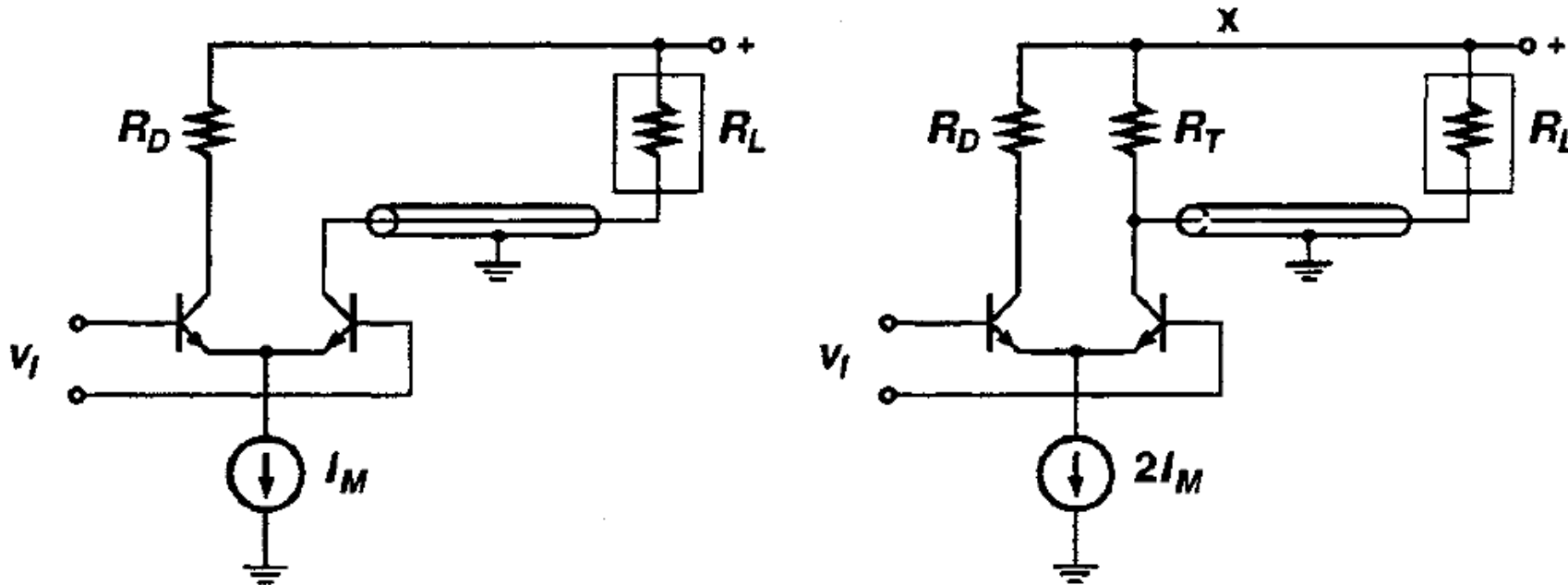
The bias current is the 0 - level current

$$I_{L,0} = I_B \quad I_{L,1} = I_B + I_M$$

The bias current is the average current
- AC-coupled case

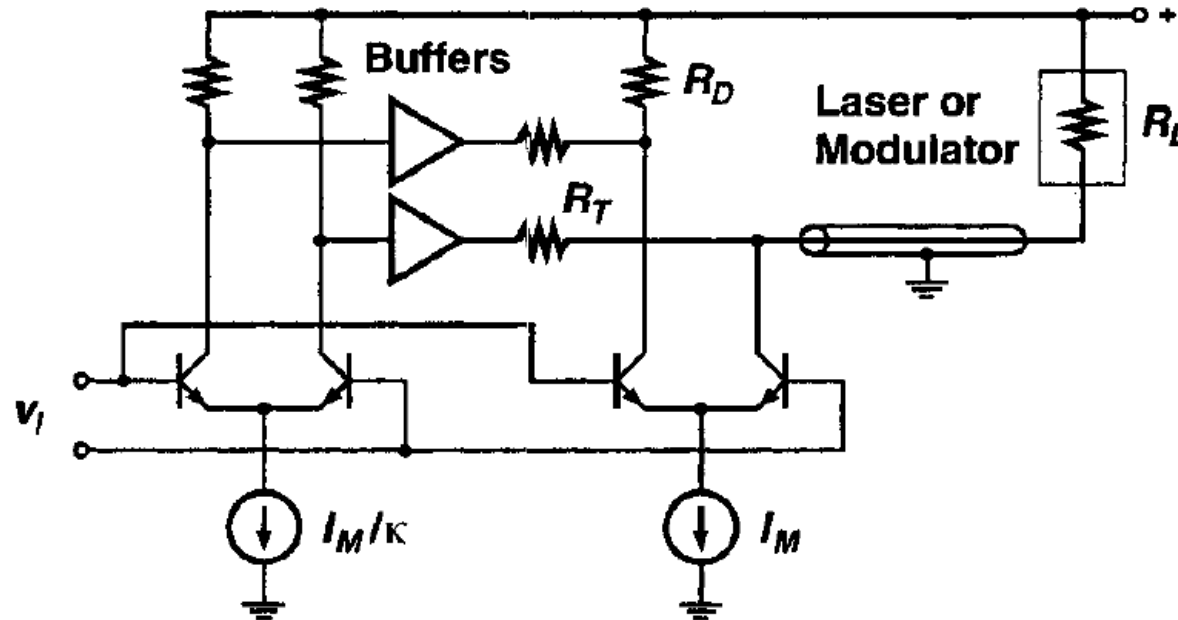
$$I_{L,0} = I_B - \frac{I_M}{2} \quad I_{L,1} = I_B + \frac{I_M}{2}$$

Termination Strategies



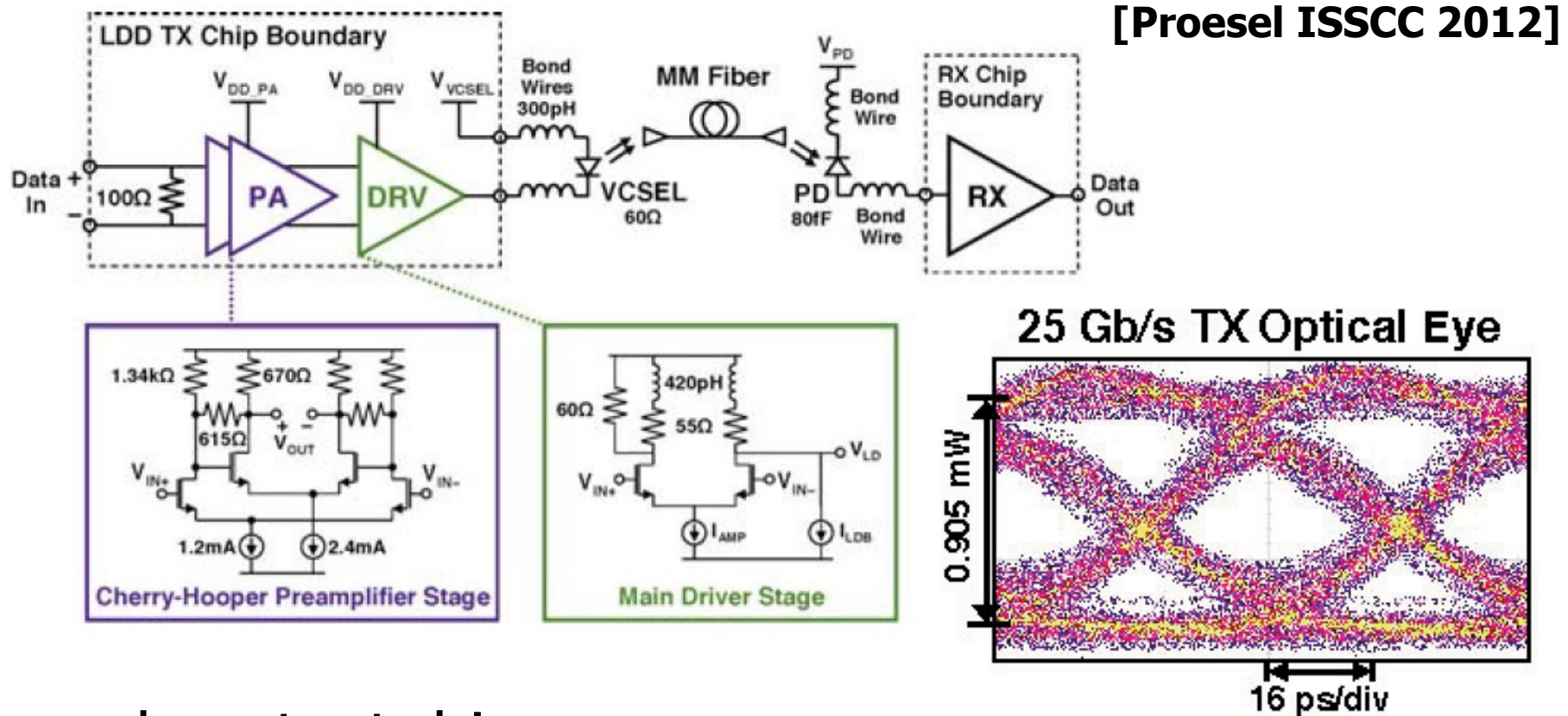
- The laser interface with the driver determines whether double or “back” termination is necessary
- Reflections from bondwires and any laser/transmission-line mismatch can degrade high-speed performance
- Driver on-die termination improves this at a power cost

Active Back Termination



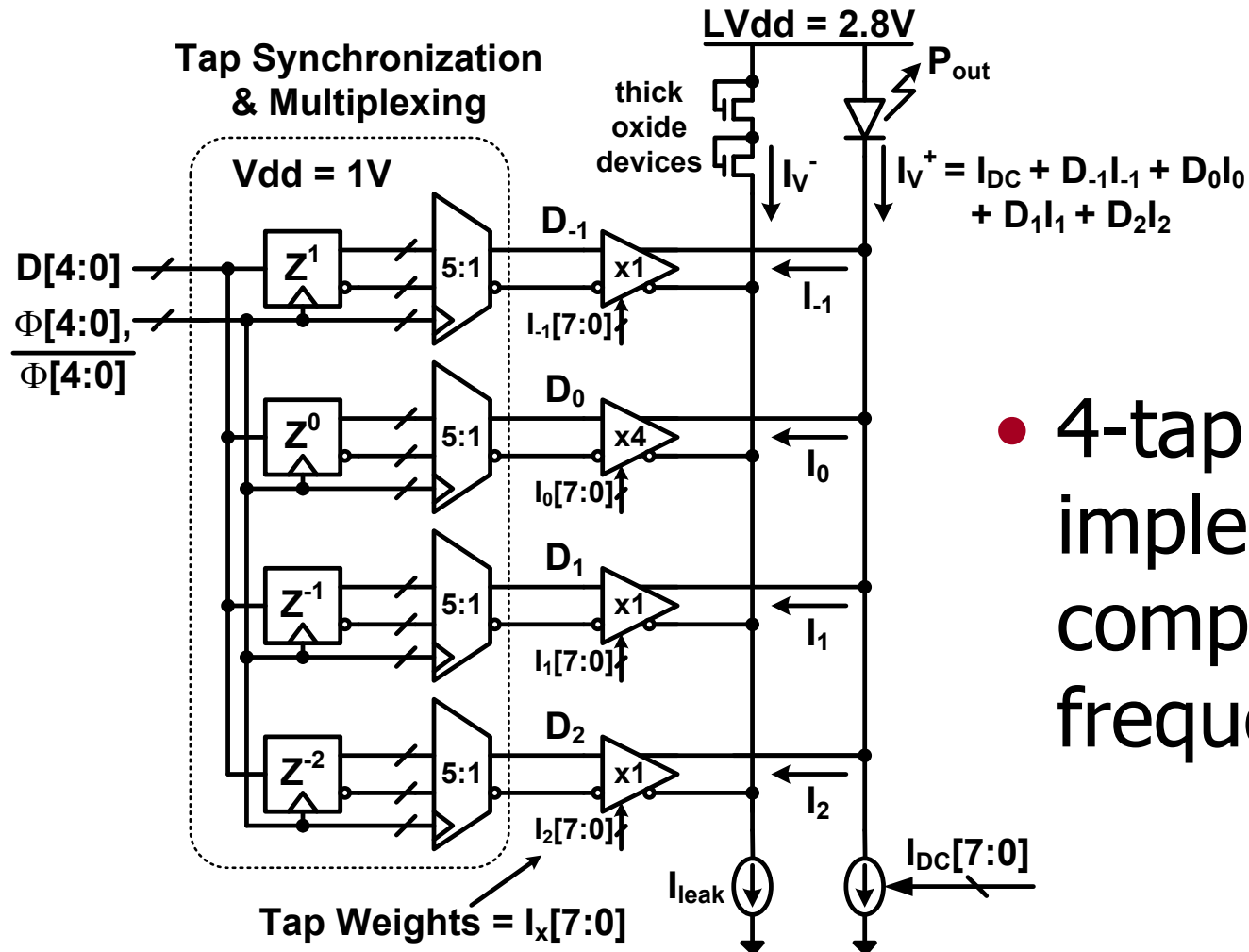
- While not a major issue for relatively low-power VCSELs, the lost current with on-die driver termination is a concern for high-power (long-haul) lasers
- This motivates the use of active back termination circuitry where the termination resistor is connected to an AC voltage generated by a replica stage
- Ideally, without reflections, no voltage drop is across R_T

25Gb/s VCSEL Link



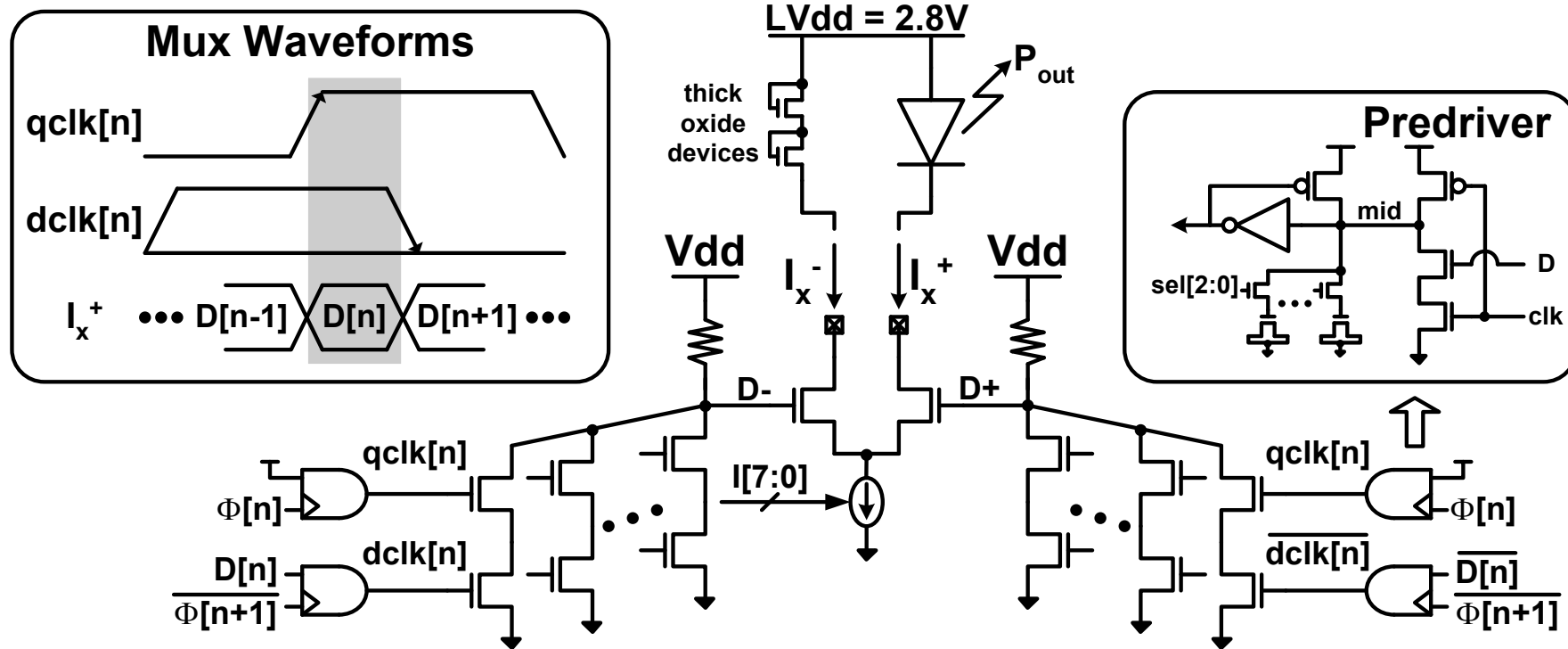
- Current-mode output driver
- Bandwidth extension achieved with on-die shunt-peaking termination in the output stage and with Cherry-Hooper preamplifier stage

Multiplexing FIR Output Driver



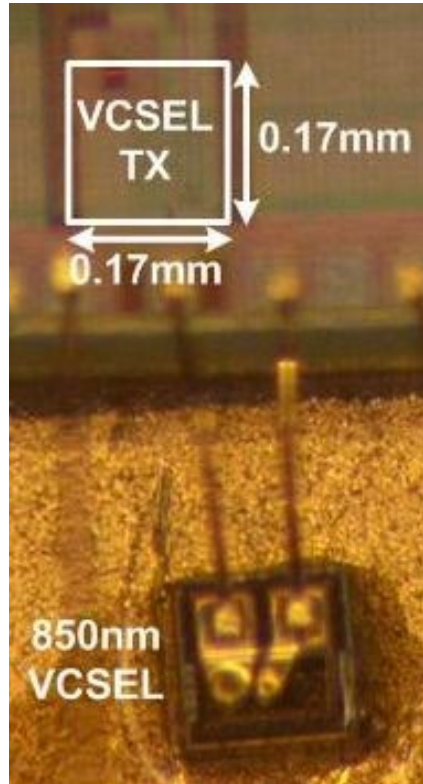
- 4-tap FIR filter implemented in driver to compensate for VCSEL frequency response

Tap Mux & Output Stage

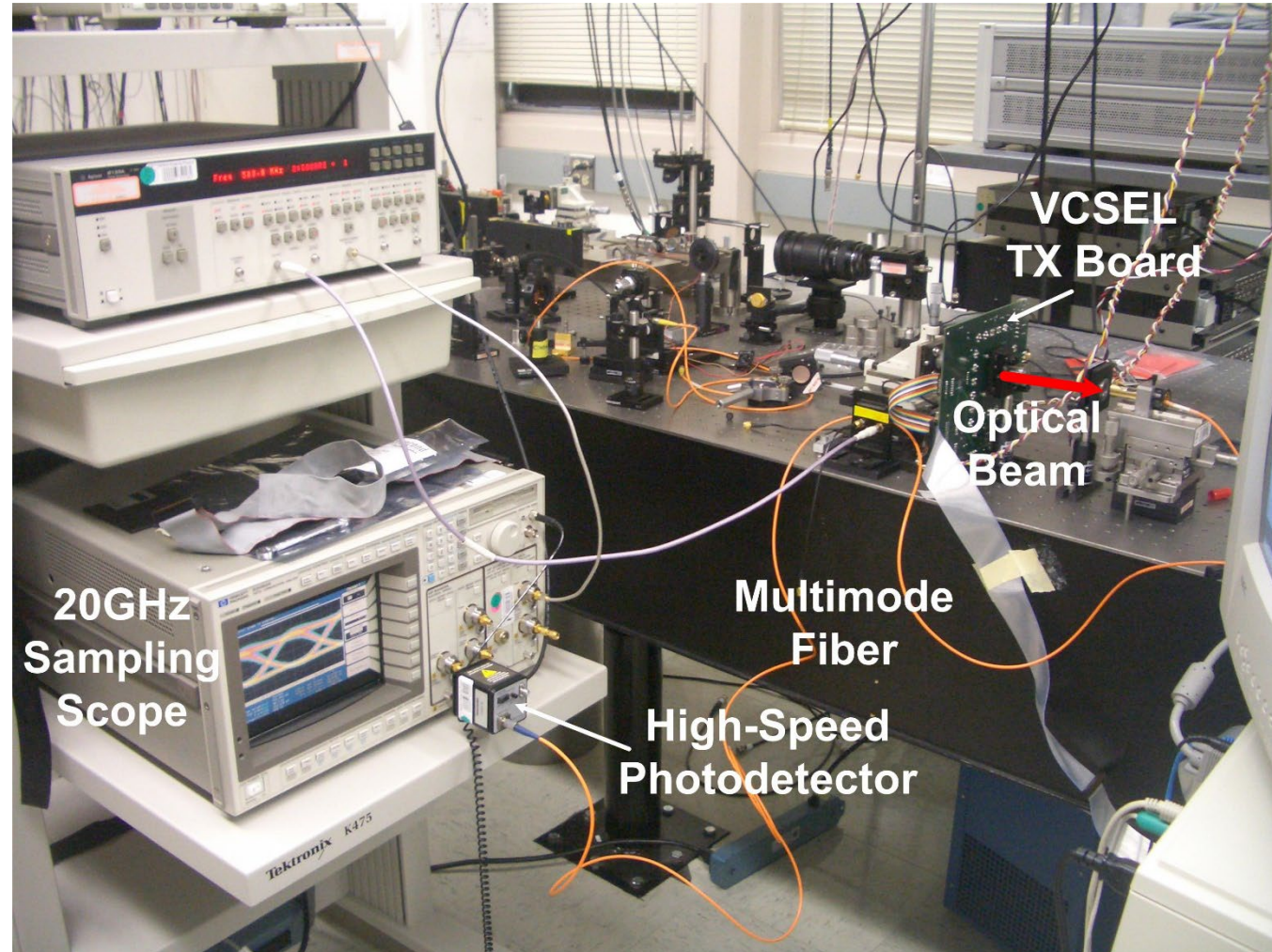


- 5:1 multiplexing predriver uses 5 pairs of complementary clock phases spaced by a bit time
- Tunable delay predriver compensates for static phase offsets and duty cycle error

VCSEL TX Optical Testing



Wirebonded
10Gb/s VCSEL



VCSEL 16Gb/s Optical Eye Diagrams

$I_{avg} = 6.2\text{mA}$, $ER = 3\text{dB}$

No Equalization →

$I_{DC} = 4.37\text{mA}$

$I_{MOD} = 3.66\text{mA}$

w/ Equalization ↘

$I_{DC} = 3.48\text{mA}$

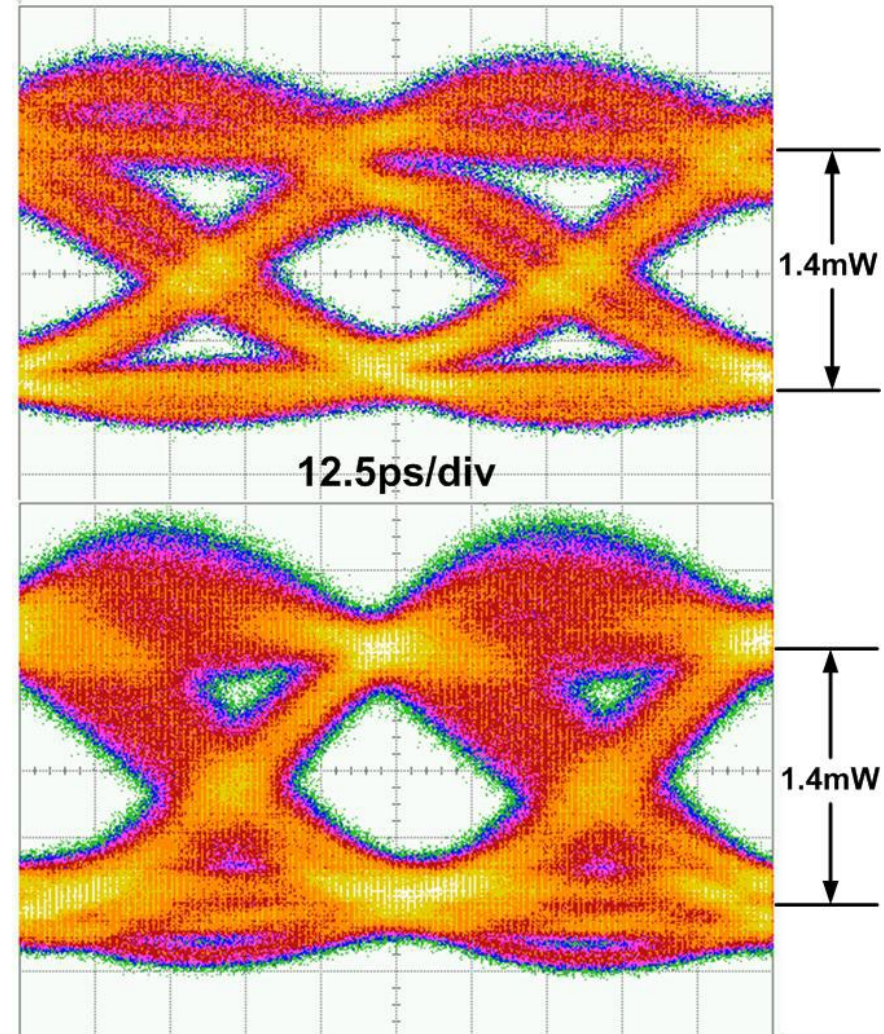
$I_{-1} = -0.70\text{mA}$

$I_0 = 4.36\text{mA}$

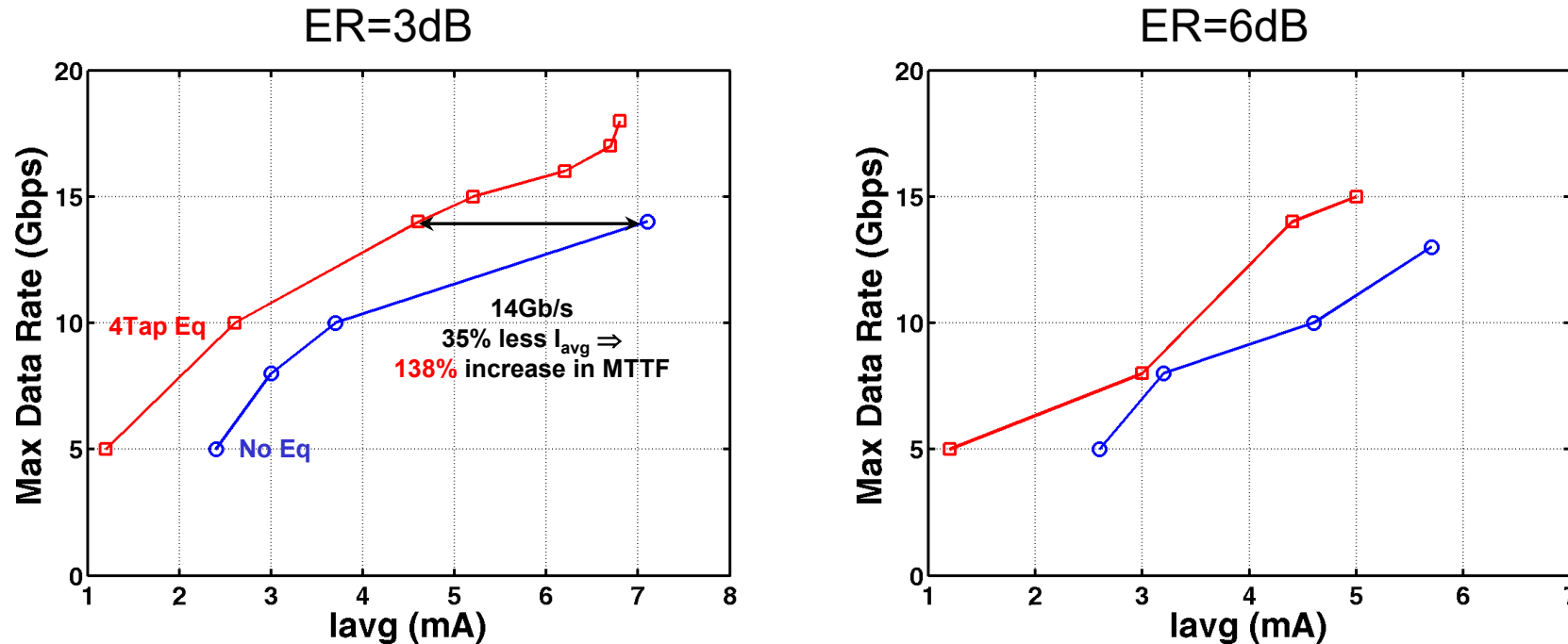
$I_1 = -0.19\text{mA}$

$I_2 = 0.19\text{mA}$

Equalization increases
vertical eye opening
45% at 16Gb/s



Equalization Performance



- Maximum data rate vs Average current
 - Min 80% eye opening & <40% overshoot
- Equalization allows lower average current for a given data rate
- Linear equalizer limited by VCSEL nonlinearity

PAM2 VCSEL Driver w/ 2-Tap Nonlinear FFE

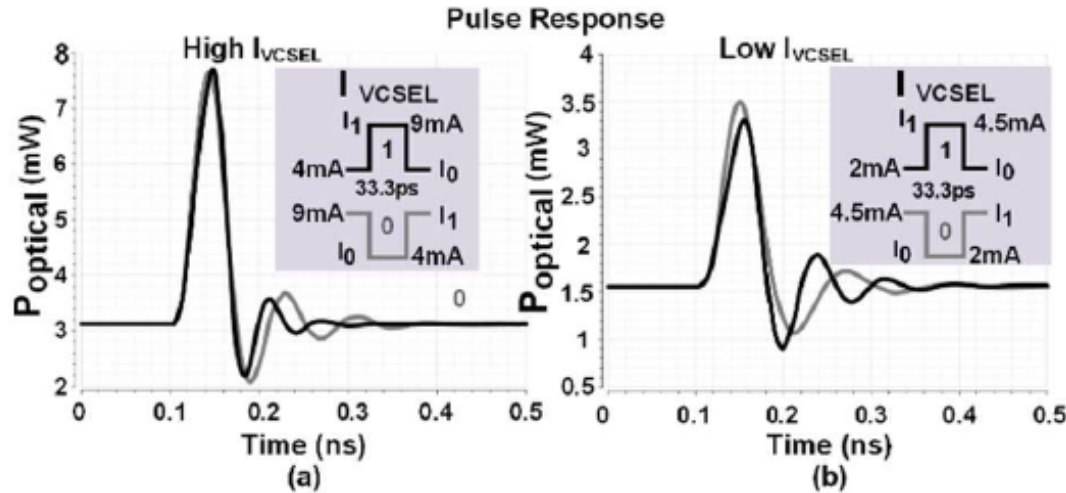


Fig. 2. VCSEL pulse responses for (a) high and (b) low I_{VCSEL} .

- VCSEL's bias-dependent frequency response results in nonlinear transient pulse responses
- A 2-tap non-linear equalizer with different equalization taps for high and low pulses provides performance improvement

[Raj CICC 2015]

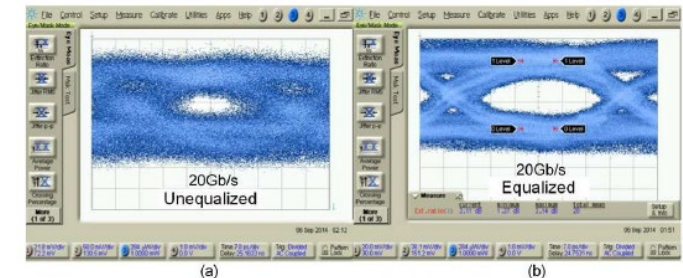
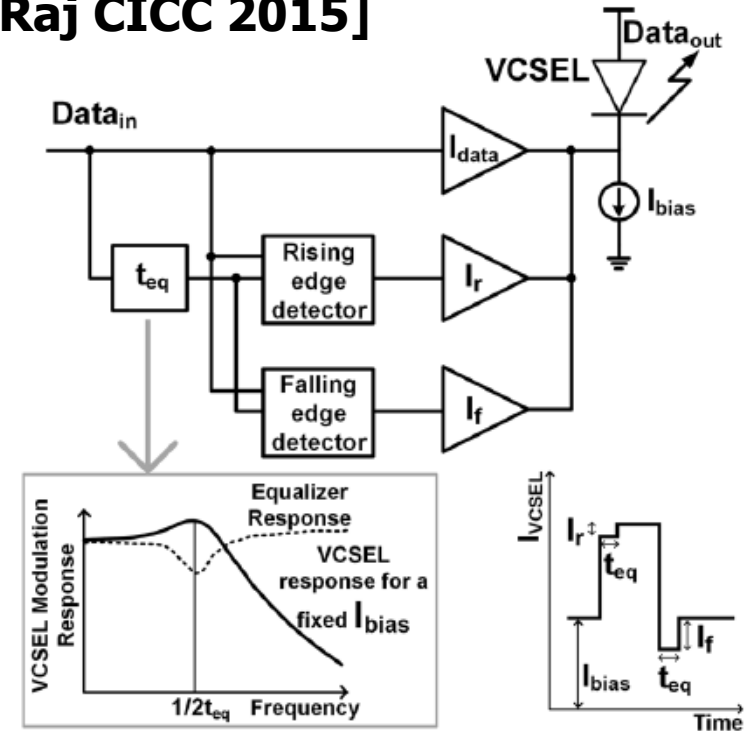
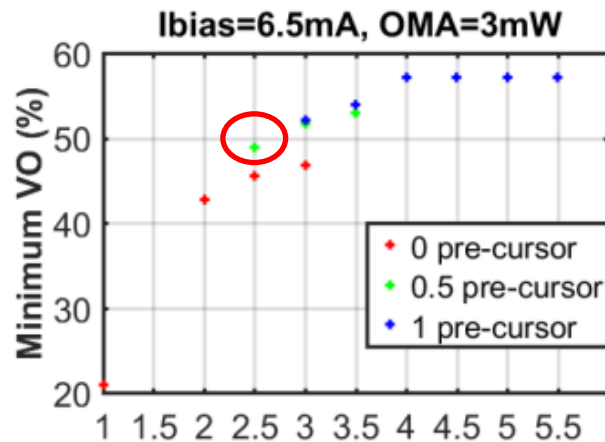
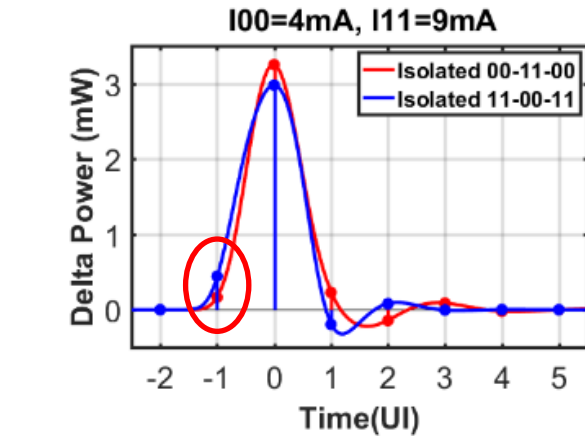
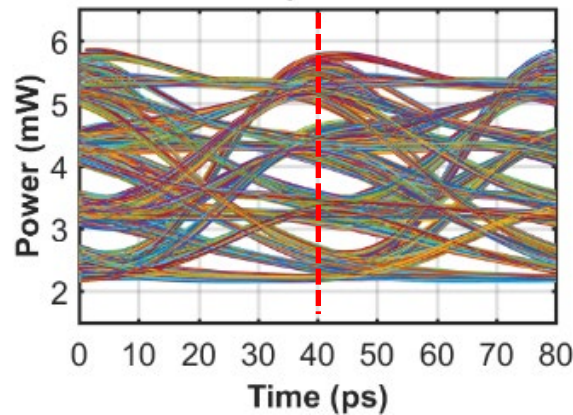
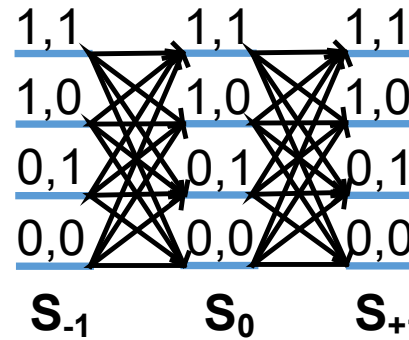


Fig. 8 Measured optical eye-diagram for PRBS-15 data at 20Gb/s. (a) Unequalized (b) Equalized.

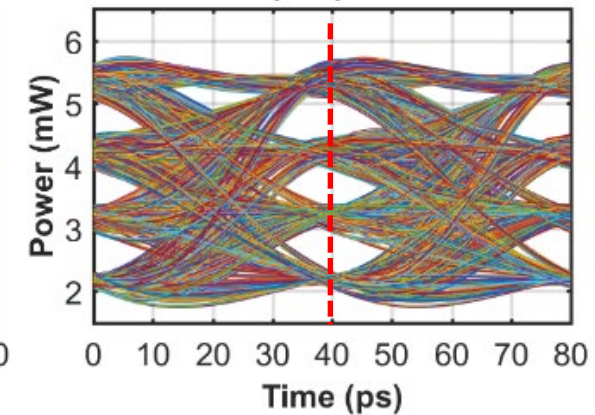
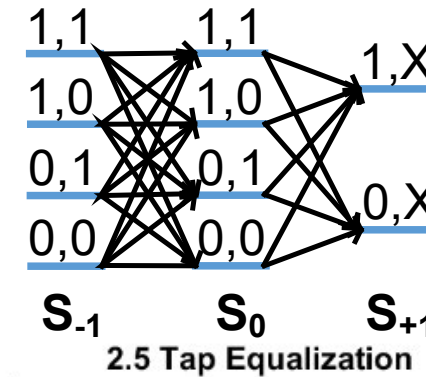
PAM4 VCSEL Driver w/ 2.5-Tap Nonlinear FFE



3-tap equalizer

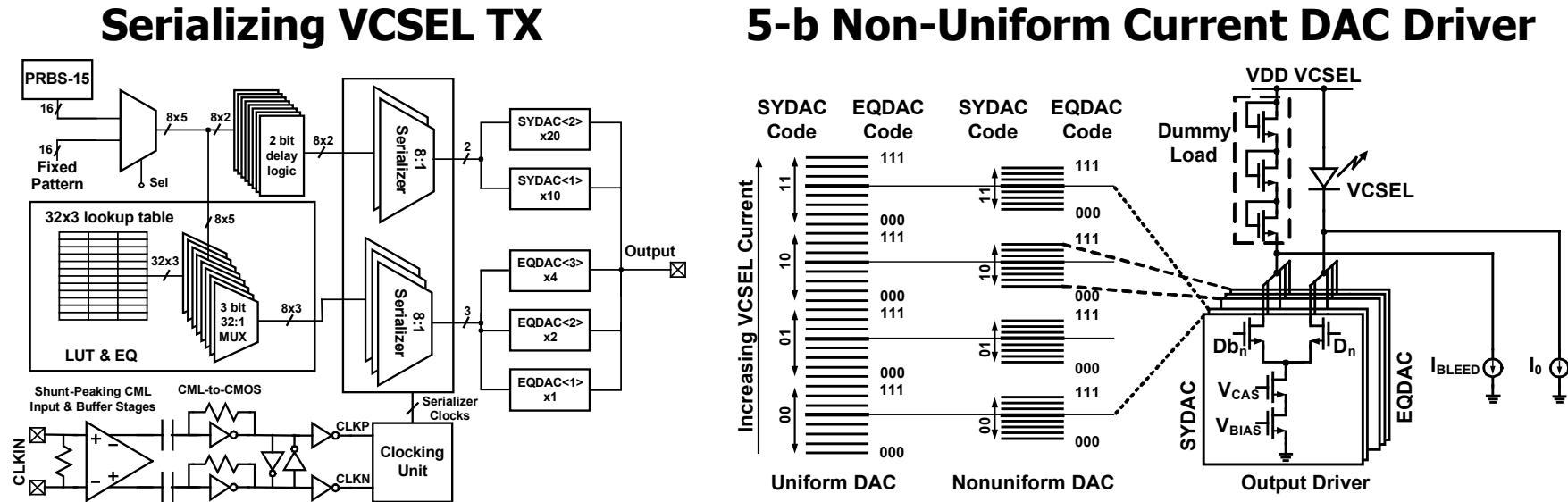


2.5-tap equalizer with the pre-cursor weight only a function of the MSB



- A 2.5-tap nonlinear equalizer, with the first pre-cursor weight only dependent on the MSB, is a good compromise between complexity and performance

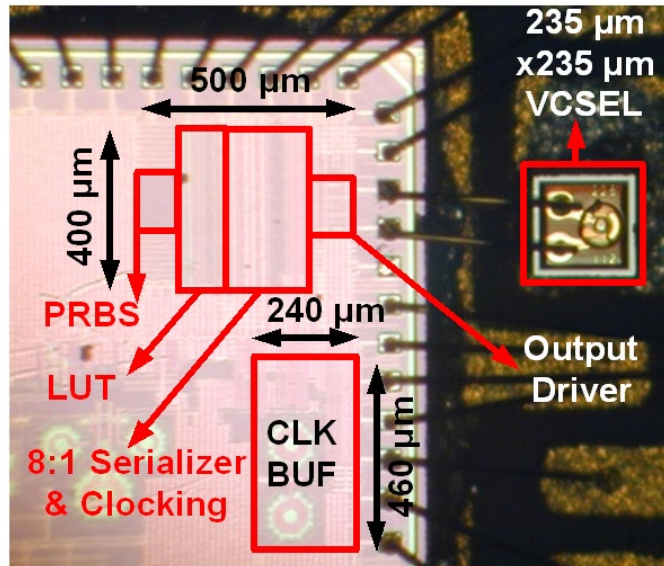
Serializing VCSEL TX & Output Stage



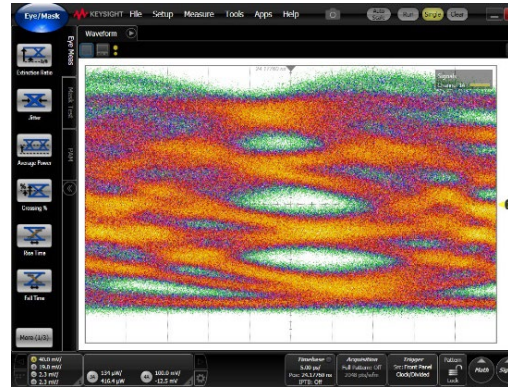
- VCSEL transmitter serializes 16 bits or 8 PAM-4 symbols
- Output stage is a 5-bit non-uniform current-mode DAC
 - MSB and MSB-1 set the main PAM-4 symbol levels
 - 3 LSB currents implement the 2.5-tap equalizer with the symbol pattern selecting the weighting from the 32X3 LUT

50Gb/s PAM4 Experimental Results

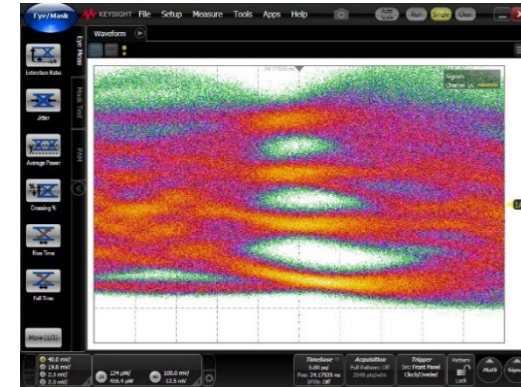
[Tyagi PTL 2018]



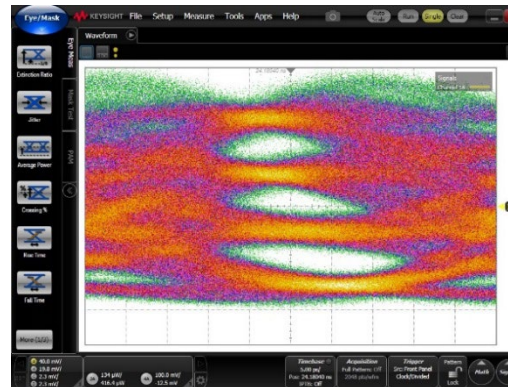
No Equalization



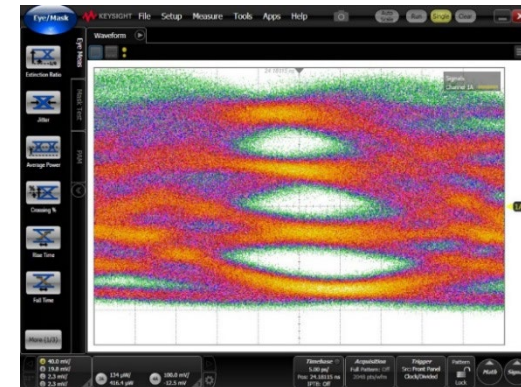
2-Tap Linear



2.5-Tap Linear

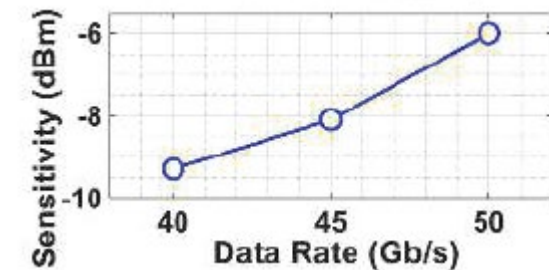
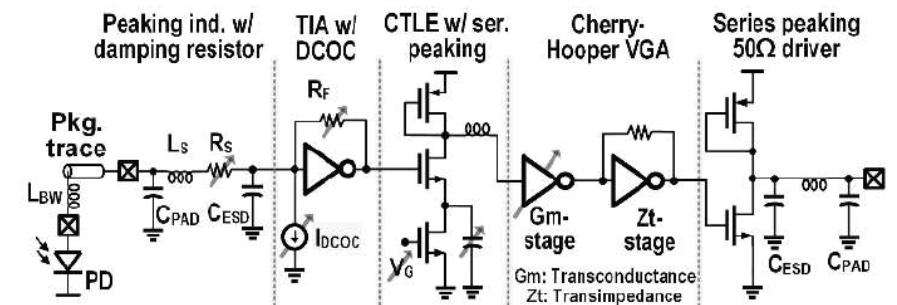
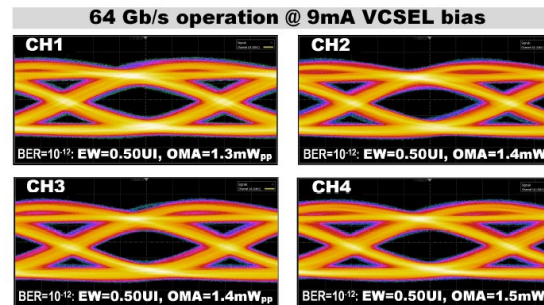
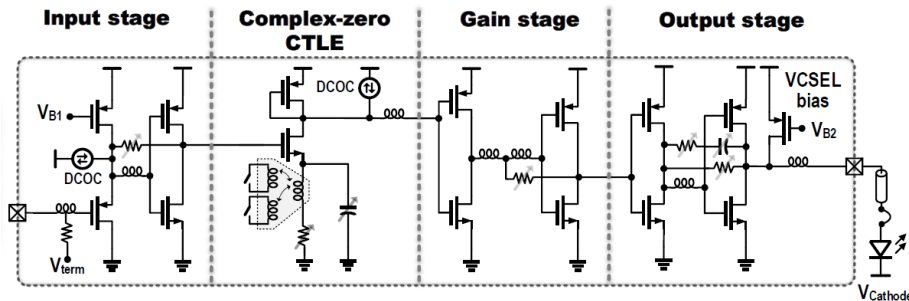
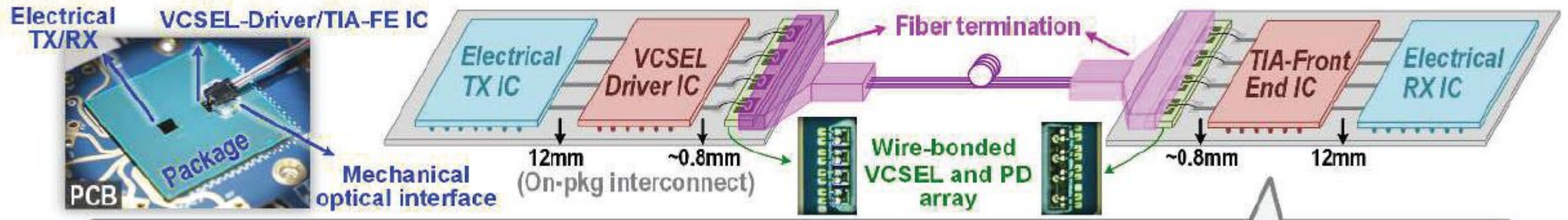


2.5-Tap Nonlinear



- Core transmitter area is 0.2mm^2
- 2.5 tap nonlinear equalizer improves eye height and timing alignment of the 3 PAM4 eyes

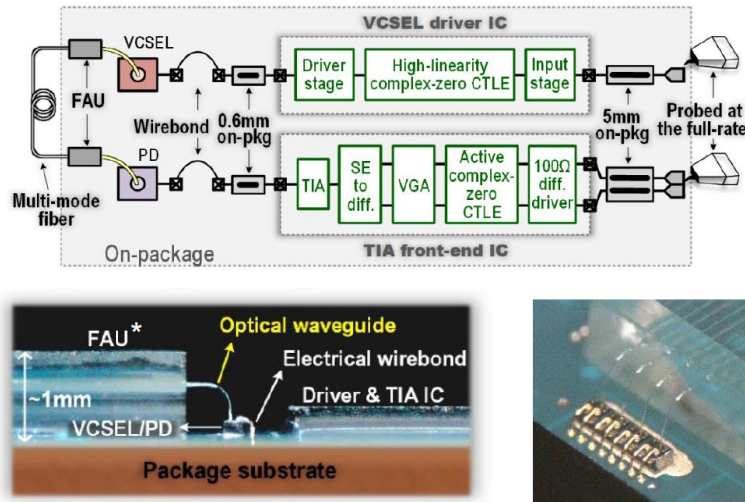
64Gb/s PAM2 VCSEL Co-Packaged Optical XCVR



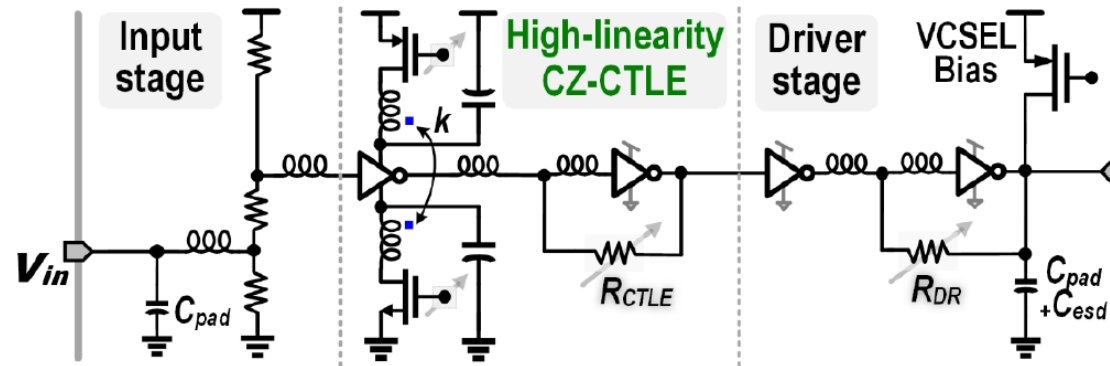
- 4-channel TX partitioned into 4:1 serializing Electrical TX IC with low-swing driver to VCSEL Driver IC
- VCSEL driver has complex-zero CTLE and Cherry-Hooper output stage
- 4-channel RX partitioned into TIA Front-End IC with common source driver to Electrical RX IC with 1:4 deserialization
- Shunt-feedback TIA cascaded with CTLE on Front-End IC followed by 2-tap FFE in Electrical RX IC allows for 50Gb/s PAM2

- TX=1.4pJ/b, RX=1.5pJ/b
- [TX: Mondal ISSCC 2024]**
[RX: Krishnamurthy VLSI 2024]

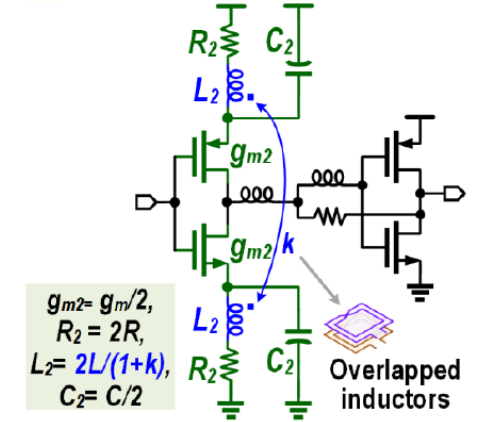
128Gb/s PAM4 VCSEL Co-Packaged TX



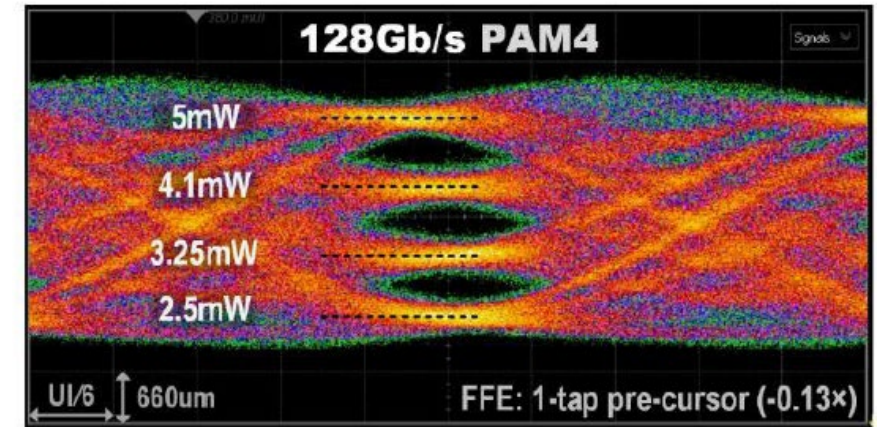
[Krishnamurthy ISSCC 2025]



D: High-linearity compact CZ-CTLE



- Single-ended VCSEL driver has high-linearity complex-zero CTLE and Cherry-Hooper output stage
- High-linearity CTLE
 - Complementary G_m improves THD and power
 - Inductor area reduced by P/N coupling
- 128Gb/s PAM4 with OMA=2.5mW and 0.31pJ/b
- Utilized photonic wirebonds between VCSEL array and fiber attach unit to lower height and reduce footprint



Next Time

- Mach-Zehnder Modulator (MZM) TX